

CSE471
Assignment 1

Name : Mihir Das

ID : 22299480

Section : 9

Department : CSE

Ans. to the Q.No.1

	2013	2014	2015
Music downloads	757500	787800	811434
Customer subscription	950000	988000	1017640
CD Sales	205000	213200	219596
Gift Cards	153000	159120	163894
Total Benefits	2065500	2148120	2212564

$$\text{Total development cost for 2012} = (42000 + 120000 + 50000 + 5000 + 2000 + 250000 + 10000 + 25000) = 279000$$

	2012	2013	2014	2015	
Total costs Benefits	0	2065500	2148120	2212564	6426184
Total costs	279000	461000	469700	483252	1692952
Net Benefit	(279000)	1604500	1678420	1729312	
Cumulative Net Benefit	(279000)	1325500	3003920	4733232	

$$\begin{aligned} \text{a) ROI} &= \frac{\text{total benefit} - \text{total cost}}{\text{total cost}} \times 100\% \\ &= \frac{6426184 - 1692952}{1692952} \times 100\% \\ &= 279.58\% \end{aligned}$$

(Ans)

$$\textcircled{b} \text{ BEP} = \left(\frac{\text{no. of years of negative cumulative cash flow}}{\text{cumulative cash flow}} \right) + \left(\frac{\text{Net benefits} - \text{cumulative Net Benefit}}{\text{Net Benefits}} \right)$$

$$= 0 + \frac{1604500 - 1325500}{1604500}$$

$$= 0.1738$$

$$\approx 0.174 \text{ years (Ans)}$$

$\textcircled{c}$	2012	2013	2014	2015	Total
Total Benefits	0	2065500	2148120	2212564	62426184
PV of Total Benefits		1948585	1911819	1857711	5718115
Total costs	279000	461000	469700	483252	
PV of total costs	279000	434905	418031	405747	1537683

Now, Net Present Value :

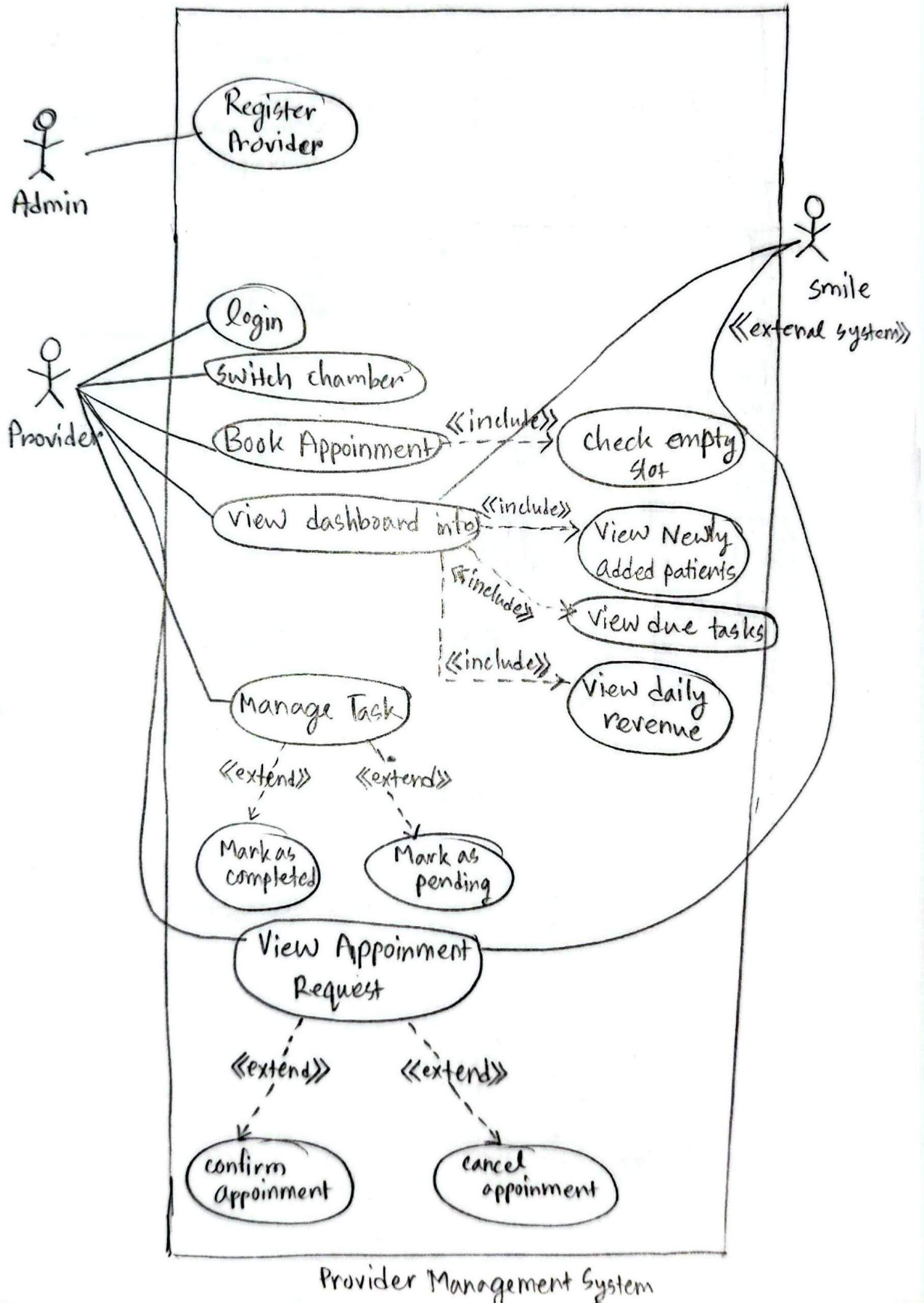
TB = Total Benefit; TC = Total cost

$$\text{NPV} = \sum \{ \text{PV}(\text{TB}) - \sum \text{PV}(\text{TC}) \}$$

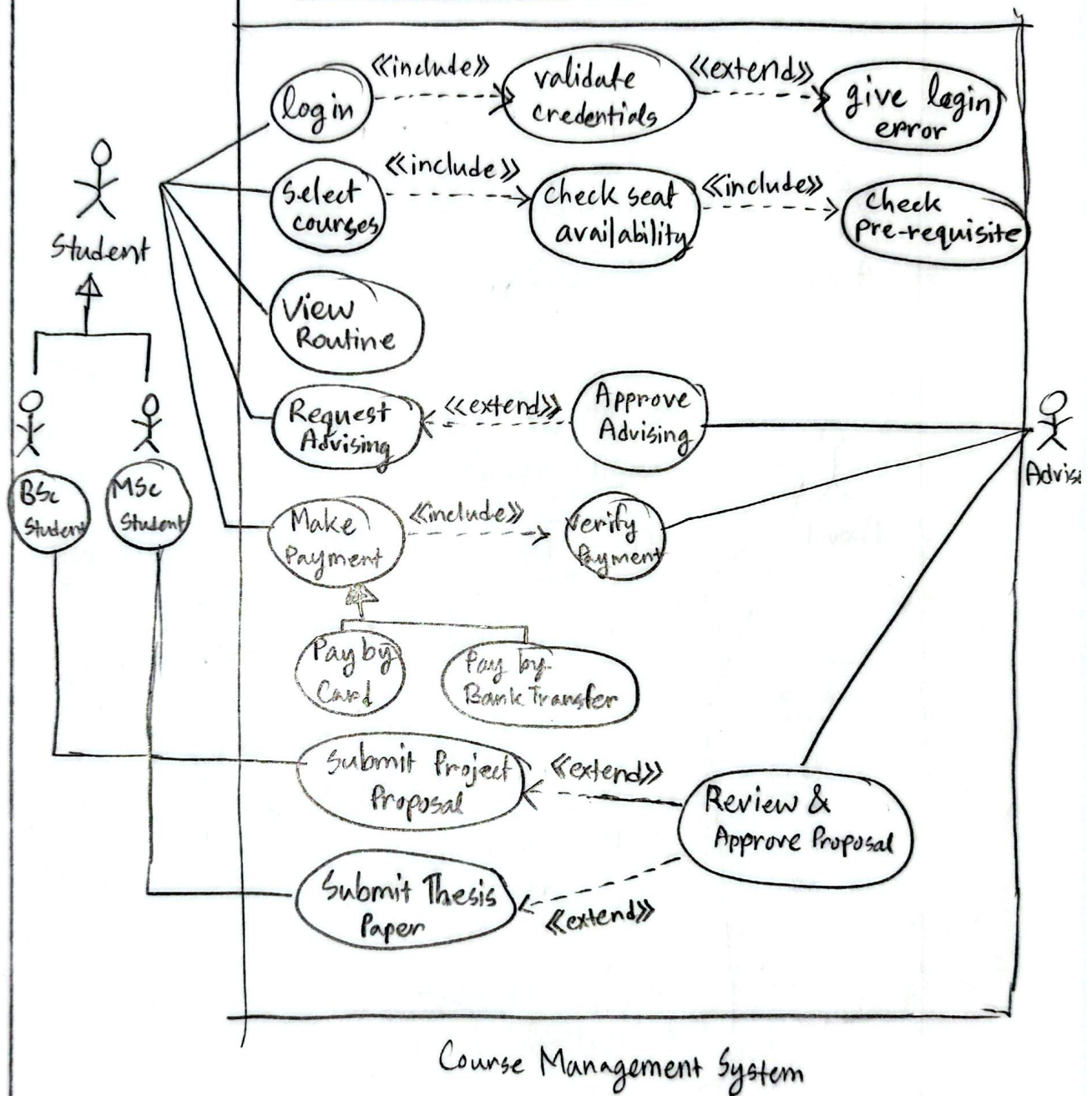
$$= 5718115 - 1537683$$

$$= 4180432 \text{ (Ans)}$$

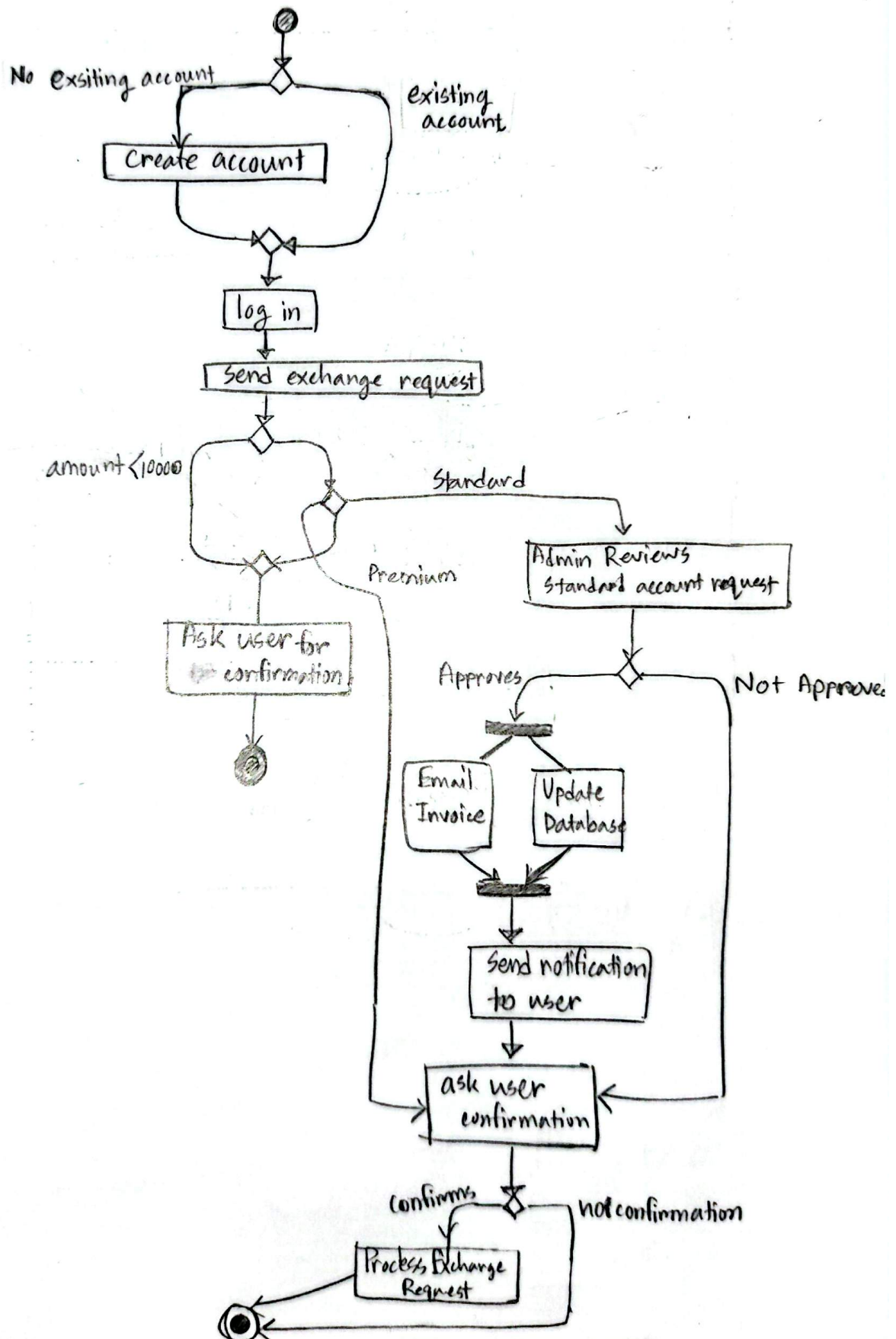
Ans. to the Q.No. 2



Ans. to the Q.No. 3



Ans. to the Q.No.4



Ans. to the QNo. 5

